CMPE 300
ANALYSIS OF ALGORITHMS

Department of Computer Engineering, Boğaziçi University

Fall 2026

Instructor: Tunga Güngör

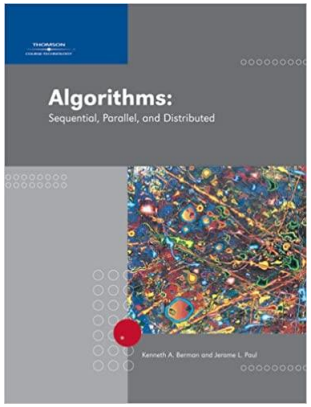
E-mail: gungort@bogazici.edu.tr

Teaching Assistant: Kübra Uykun

E-mail: kubra.uykun@bogazici.edu.tr

Course Overview – Textbook

- **Textbook**
  - Kenneth A. Berman and Jerome L. Paul, *Algorithms: Sequential, Parallel, and Distributed*, Thomson Pub., 2005



- **Reference book**
  - Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein, *Introduction to Algorithms*, 3rd ed., MIT Press, 2009

Course Overview – Course Schedule

- Introduction
- Algorithm complexity (Best-case, Worst-case, Average complexity)
- Asymptotic analysis (Growth rate, Asymptotic notation, Comparison of growth rates)
- Analysis of example algorithms
- Interpolation (θ-invariant under scaling, Scale invariant classes)
- Stable, in-place, on-line algorithms
- Adjacent-key comparison-based algorithms
- Recurrence relations (Forward substitution, Backward substitution, Change of variable)
- Master Theorem
- Divide-and-conquer and graph algorithms (Fast matrix multiplication, Depth-first and breadth-first search and traversal)
- Dynamic programming and greedy method (Longest common subsequence, Knapsack problem)
- Parallel architectures (Flynn's taxonomy, Shared memory, Distributed memory)
- Parallel algorithms (Analysis, Goodness measures, Evaluation)
- Lower bound theory (Optimality, Simple counting, Enumeration, Adversary arguments, Decision trees, Reduction)
- Probabilistic algorithms (Randomizing deterministic algorithms, Monte Carlo algorithms, Las Vegas algorithms)

Course Overview – Evaluation

- Evaluation (subject to change)
  - Quizzes : 15%
  - Projects (3) : 40% (13%, 14%, 13%)
  - Midterm : 20%
  - Final : 25%

Course web page: https://moodle.bogazici.edu.tr

Course Overview – Notes

- Midterm will be held during the week 2-6.11.2026.
- The midterm and final examinations will be “closed books and notes”.
- You can follow the announcements via the university’s Moodle system (https://moodle.bogazici.edu.tr).

  (Students registered to the course are automatically added to the course in Moodle.)
- The textbook is available in Moodle.
- There will be short, unscheduled quizzes throughout the semester.
- *Attendance to both the midterm exam and the final exam, attendance to at least three of the quizzes, submitting all the projects, and getting a grade of at least 10 (out of 100) in both the midterm exam and the final exam are mandatory. Otherwise, you will fail the course, regardless of the grades obtained in other parts of the course.*